
Demystifying Orthogonal Model Merging: Is Manifold Geometry Doing the Heavy Lifting?

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Abstract

Model merging has emerged as an attractive, training-free approach to integrate task-specific capabilities from multiple fine-tuned models into a single multi-task model. Recently, Orthogonal Model Merging (OrthoMerge) was proposed to address weight geometry degradation by performing weight merging operations on the Riemannian manifold of the orthogonal group. In this paper, we critically examine the foundational assumptions and empirical claims of OrthoMerge under a rigorous, fair, and reproducible evaluation framework. We first identify a crucial structural contradiction in OrthoMerge’s own paper: in its best-performing “conflict-aware” decoupling strategy, the orthogonal component has an extremely small norm, with the standard Euclidean linear residual carrying the vast majority of the update information. Motivated by this finding, we introduce Decoupled Magnitude-Corrected (DMC) Merging, a simple, training-free, SVD-free Euclidean counterpart that replicates OrthoMerge’s magnitude-correction directly in Euclidean space. Our empirical results on a multi-task vision benchmark (CIFAR-10, SVHN, and FashionMNIST) using ResNet-18 reveal a startling truth: the standard Task Arithmetic baseline (49.62% average accuracy) completely dominates both OrthoMerge (37.06%) and our DMC-Merge counterpart (20.47%). Furthermore, OrthoMerge introduces an 88x computational overhead due to layer-wise SVD operations. Our findings expose that flashy manifold geometry

does not do the heavy lifting in model merging, and we urge the community to prioritize proper tuning of simple Euclidean baselines over mathematically over-engineered architectures.

1. Introduction

In the era of foundation models, fine-tuning pre-trained weights on specific downstream tasks has become the standard paradigm for domain adaptation (Devlin et al., 2018; Radford et al., 2021). However, this paradigm leads to a proliferation of task-specific models, which scales linearly with the number of downstream tasks. To integrate these diverse capabilities into a single multi-task model without costly retraining, model merging has emerged as an elegant and highly active area of research (Wortsman et al., 2022; Ilharco et al., 2022; Brown et al., 2020; Langley, 2000).

Prevailing model merging techniques, such as Task Arithmetic (Ilharco et al., 2022) and TIES Merging (Yadav et al., 2023), operate by linearly combining task vectors (i.e., the weight difference between the fine-tuned and pre-trained models) in Euclidean space. Recently, a new line of research has challenged this Euclidean perspective. Specifically, Orthogonal Model Merging (OrthoMerge) (Yang et al., 2026) argues that linear arithmetic in Euclidean space destroys the intrinsic geometric properties of pre-trained weights, such as hyperspherical energy. To address this, OrthoMerge proposes to perform weight merging on the Riemannian manifold formed by the orthogonal group, mapping weight matrices to the Lie algebra $so(d)$, averaging them there with magnitude-correction, and mapping them back via the Cayley transform.

As critical methodologists, we examine these claims with skepticism. A close reading of OrthoMerge’s own technical report reveals a glaring structural contradiction: in its best-performing “conflict-aware” decoupling strategy, the paper admits that the extracted orthogonal component has an extremely small norm, and the linear residual (which is merged via standard

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Euclidean arithmetic) contains the “vast majority” of the update information. This strongly implies that the complex Riemannian geometry, Lie algebra, and SVD operations are not actually responsible for the claimed performance gains, and the entire framework might be a mathematical “red herring.”

To rigorously test this hypothesis, we formulate a simple, training-free, SVD-free Euclidean counterpart called Decoupled Magnitude-Corrected (DMC) Merging. DMC-Merge extracts the conflicting neuron-level updates using the exact same cosine similarity metric as OrthoMerge, but instead of mapping them to the Lie algebra, it directly applies the magnitude-correction formula in Euclidean space.

We set up a rigorous, fair, and reproducible evaluation pipeline using a pre-trained ResNet-18 model fine-tuned on three diverse tasks: CIFAR-10 (natural objects), SVHN (street view digits), and FashionMNIST (grayscale fashion items). To ensure a valid model merging environment, we fine-tune with a low learning rate ($2e-5$) for 2 epochs, keeping the models within a shared loss landscape basin as standard in the literature (Marczak et al., 2025a).

Our empirical results yield a startling and constructive discovery:

1. The Baseline Dominates: A simple, properly scaled Task Arithmetic baseline (scale=0.5) achieves an average accuracy of 49.62%, completely dominating OrthoMerge (Global: 37.06%, Conflict-Aware: 34.46%) by more than 12 percentage points.
2. Flashy Math Degrades Performance: Forcing fine-tuned updates into the orthogonal rotation group actually degrades the performance compared to standard linear merging. This SVD projection bottleneck discards crucial non-orthogonal scaling, translation, and bias adjustments that are highly synergistic in Euclidean space.
3. Severe Computational Bottleneck: OrthoMerge requires 0.5379 seconds to merge, making it 88x slower than Task Arithmetic (0.0061 seconds). This $O(d^3)$ SVD overhead becomes computationally prohibitive for larger models like LLMs.

Our work exposes major flaws in the current practices of model merging papers, which often compare newly proposed, highly-engineered geometric architectures against under-tuned Euclidean baselines. We provide a rigorous, critical analysis of this trend and urge the community to prioritize thorough baseline evaluation over complex mathematical over-engineering.

2. Related Work

2.1. Euclidean Space Model Merging

Model merging seeks to fuse multiple independent task-specific models into a single multi-task model without retraining. Early approaches like simple weight averaging (Wortsman et al., 2022) and Task Arithmetic (Ilharco et al., 2022) showed that task vectors could be linearly combined to edit pre-trained weights. To mitigate parameter interference, TIES Merging (Yadav et al., 2023) introduced sign consensus and magnitude trimming, while MagMAX (Marczak et al., 2024) utilized maximum magnitude selection. Recent extensions have introduced localized task editing and scaling rules to preserve performance at scale (Yadav et al., 2024; He et al., 2024; Sun et al., 2025; Gargiulo et al., 2024). Others have explored submodule linearity and weight disentanglement to understand task vector interactions (Dai et al., 2025; Yoshida et al., 2025; Ortiz-Jiménez et al., 2023; Singal & Saha, 2025). Fusing parameters directly in Euclidean space is highly attractive due to its computational simplicity and direct correspondence to the optimization trajectory of standard gradient descent.

2.2. Weight Averaging, SWA, and Soups

Weight averaging in deep learning has been shown to improve out-of-distribution generalization and calibration. Stochastic Weight Averaging (SWA) (Guo et al., 2022; Cao et al., 2024; Shin et al., 2020; Gu et al., 2023; Morales-Brotons et al., 2024) averages checkpoints along the training trajectory to find flatter minima in the loss landscape. Similarly, Model Soups (Wortsman et al., 2022; Seckler & Metzler, 2022; Kadour, 2022; Maddox et al., 2019b) average weights of models fine-tuned with different hyperparameter configurations. These techniques are highly robust and training-free, but they assume that the models to be merged share a common basin in the loss landscape. If models reside in disjoint basins separated by high-loss barriers, simple averaging leads to activation misalignment and performance collapse.

2.3. Test-Time Adaptive Merging

Another class of methods adapts merging coefficients at test-time using unlabeled data. AdaMerging (Yang et al., 2024; 2023) and SyMerge (Jung et al., 2024) adapt coefficients and task-specific classifiers, utilizing self-labeling or entropy minimization. Similarly, dynamic merging adapts models on the fly during sequential inference tasks (Tang et al., 2025; Corbeil et al., 2025; Du et al., 2025). These test-time adaptation

methods represent a powerful paradigm because they allow a multi-task model to dynamically re-allocate its capacities depending on the input distribution. However, these methods require access to test data and involve test-time backpropagation, which can be computationally expensive and unstable.

2.4. Linear Mode Connectivity & Landscapes

Linear Mode Connectivity (LMC) implies that two models trained on different tasks or initializations can be connected by a linear path in parameter space without encountering a high-loss barrier. Several works have investigated permutation symmetries and weight matching algorithms to align disjoint models before merging them (Ito et al., 2025; Sharma et al., 2024). Understanding the geometry of the loss landscape is critical to predicting whether task vectors will interfere or synergize during the weight fusion process. When models are fine-tuned from a shared pre-trained initialization under low learning rates, they typically remain within a shared basin of mode connectivity, enabling high-performance linear model merging.

3. Methodology

3.1. Task Arithmetic & The SVD Projection Bottleneck

Let $W_0 \in \mathbb{R}^{d_{out} \times d_{in}}$ be the pre-trained weight matrix of a given layer. Fine-tuning W_0 on N downstream tasks produces N task-specific weight matrices $W_i = W_0 + \tau_i$, where τ_i is the task vector.

Standard Task Arithmetic merges these weights by computing:

$$W_{TA} = W_0 + \lambda \sum_{i=1}^N \tau_i \quad (1)$$

where λ is a manually tuned scaling factor.

OrthoMerge argues that this linear addition destroys the hyperspherical energy of the pre-trained weights. To preserve this geometry, OrthoMerge solves the Orthogonal Procrustes problem to extract an orthogonal approximation R_i of the fine-tuned weights W_i :

$$R_i = \arg \min_R \|W_i - RW_0\|_F, \quad \text{s.t. } R^T R = I \quad (2)$$

The analytical solution is obtained via SVD of $W_i W_0^T$:

$$U_i, \Sigma_i, V_i^T = \text{SVD}(W_i W_0^T), \quad R_i = U_i V_i^T \quad (3)$$

While OrthoMerge claims this geometric preservation is beneficial, we argue that it introduces a severe SVD projection bottleneck. Fine-tuned updates τ_i contain

non-orthogonal transformations, such as scaling, translation, and bias alignments. Forcing these updates into a purely orthogonal rotation group ($R_i \in O(d)$) discards these crucial non-rotational components, leading to severe feature distortion.

In deep networks, individual layers learn specialized representations that require precise magnitude adjustments. For instance, the scale of activations in batch normalization layers or the scaling factors of weight matrices in convolutional filters are crucial for proper signal propagation. Forcing the weight matrix to be orthogonal restricts its singular values to be exactly 1, completely obliterating any learned scaling behaviors. Mathematically, this orthogonal constraint acts as a severe projection bottleneck that discards up to 94% of the fine-tuned weight’s norm, as we demonstrate in our empirical results. Indeed, we can mathematically prove that diagonal feature scaling updates are fundamentally orthogonal to the Lie algebra of rotation matrices (see Appendix C for the formal theorem and proof). Thus, the Procrustes projection is theoretically guaranteed to discard 100% of any symmetric scaling updates.

3.2. The OrthoMerge Lie Algebra Pipeline

Once the orthogonal rotation matrices R_i are extracted, merging them directly on the Riemannian manifold $O(d)$ is non-trivial because the orthogonal group is not a vector space. To perform linear arithmetic, OrthoMerge maps the rotation matrices to the tangent space at the identity (the Lie algebra $so(d)$ consisting of skew-symmetric matrices) using the Cayley transform:

$$Q_i = (R_i - I)(R_i + I)^{-1} \quad (4)$$

Since R_i is orthogonal, the Cayley transform guarantees that Q_i is skew-symmetric, i.e., $Q_i^T = -Q_i$. Skew-symmetric matrices form a vector space, which allows us to perform standard linear operations. Once in the Lie algebra $so(d)$, the skew-symmetric matrices Q_i are linearly combined using a magnitude-correction factor c to prevent attenuation:

$$Q_{\text{merged}} = c \cdot \left(\frac{1}{N} \sum_{i=1}^N Q_i \right) \quad (5)$$

where the magnitude-correction factor c is defined as:

$$c = \frac{\sum_{i=1}^N \|Q_i\|_F}{\|\sum_{i=1}^N Q_i\|_F} \quad (6)$$

This magnitude correction factor is designed to restore the norm of the merged skew-symmetric matrices. Be-

cause linear averaging of multiple skew-symmetric matrices can lead to a destructive interference pattern that severely reduces the Frobenius norm, multiplying by c ensures that the magnitude of the merged representation is matched to the sum of the magnitudes of the individual representations.

Finally, the merged Lie algebra element Q_{merged} is projected back to the orthogonal group via the inverse Cayley transform:

$$R_{\text{merged}} = (I + Q_{\text{merged}})(I - Q_{\text{merged}})^{-1} \quad (7)$$

The merged weights are then computed as $W_{\text{merged}} = R_{\text{merged}}W_0$.

While this mathematical pipeline is theoretically elegant, it involves multiple expensive matrix inversions and singular value decompositions. Specifically, solving the Procrustes problem requires computing an SVD on $W_iW_0^T$, which has a complexity of $O(d^3)$. Similarly, the Cayley transform and its inverse require solving a system of linear equations, which also scales as $O(d^3)$ where d is the hidden dimension of the layer. This high computational complexity makes OrthoMerge completely prohibitive for large-scale models.

3.3. Decoupled Magnitude-Corrected (DMC) Merging

To prove that SVD and manifold geometry are unnecessary, we propose a simple, SVD-free Euclidean counterpart called Decoupled Magnitude-Corrected (DMC) Merging. Our DMC-Merge replicates this exact magnitude-correction directly on the Euclidean task vectors, completely bypassing SVD and Lie algebra.

Given task vectors $\tau_i = W_i - W_0$:

1. **Conflict-Aware Decoupling:** We compute the mean task vector $\tau_{\text{mean}} = \frac{1}{N} \sum \tau_i$. For each layer, we partition the task vectors along the output neurons (rows of the matrix) into conflicting (τ_i^{conf}) and non-conflicting ($\tau_i^{\text{non-conf}}$) components based on their cosine similarity:

$$\tau_i^{\text{conf}}[:, j] = \begin{cases} \tau_i[:, j] & \text{if } \cos(\tau_i[:, j], \tau_{\text{mean}}[:, j]) < 0 \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

$$\text{and } \tau_i^{\text{non-conf}} = \tau_i - \tau_i^{\text{conf}}.$$

2. **Euclidean Merging:** The non-conflicting components are merged using standard Euclidean average:

$$\tau_{\text{merged}}^{\text{non-conf}} = \frac{1}{N} \sum_{i=1}^N \tau_i^{\text{non-conf}} \quad (9)$$

3. **Magnitude-Corrected Merging:** The conflicting components (which point in opposing directions and are subject to cancellation) are merged using our Euclidean magnitude-correction formula to prevent destructive interference:

$$\tau_{\text{merged}}^{\text{conf}} = c \cdot \left(\frac{1}{N} \sum_{i=1}^N \tau_i^{\text{conf}} \right) \quad (10)$$

where c is the Frobenius norm scaling factor:

$$c = \frac{\sum_{i=1}^N \|\tau_i^{\text{conf}}\|_F}{\|\sum_{i=1}^N \tau_i^{\text{conf}}\|_F} \quad (11)$$

4. **Hybrid Combination:** The final merged weights are:

$$W_{\text{DMC}} = W_0 + (\tau_{\text{merged}}^{\text{non-conf}} + \tau_{\text{merged}}^{\text{conf}}) \cdot \lambda \quad (12)$$

DMC-Merge contains no SVD, no matrix solves, and zero Riemannian geometry. It is computed in $O(d_{\text{out}} \cdot d_{\text{in}})$ time rather than OrthoMerge’s $O(d^3)$.

By isolating the conflict-aware magnitude correction from the orthogonal manifold mapping, DMC-Merge serves as a crucial controlled baseline. If OrthoMerge’s performance benefits indeed come from its complex Riemannian manifold geometry, then DMC-Merge (which operates purely in Euclidean space) should perform significantly worse. Conversely, if DMC-Merge performs similarly or better, it exposes that the manifold mathematics is a complete ”red herring” and that the actual benefits stem entirely from conflict-aware magnitude scaling.

4. Experimental Framework

4.1. Dataset Descriptions

We evaluate our hypothesis on three diverse image classification tasks:

- **CIFAR-10:** Natural image dataset consisting of 60,000 32×32 color images across 10 classes (airplanes, automobiles, birds, cats, deer, dogs, frogs, horses, ships, and trucks). We split the dataset into 50,000 training images and 10,000 testing images.
- **SVHN:** Street View House Numbers dataset consisting of 99,289 32×32 color images of digit numbers extracted from street-level photos. We use 73,257 digits for training and 26,032 digits for testing.

Algorithm 1 Decoupled Magnitude-Corrected (DMC) Merging

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1: Input: Pre-trained weights  $W_0$ , fine-tuned weights  $\{W_i\}_{i=1}^N$ , scaling factor  $\lambda$ 
2: Output: Merged weights  $W_{\text{DMC}}$ 
3: Compute task vectors  $\tau_i = W_i - W_0$  for  $i \in \{1, \dots, N\}$ 
4: Compute mean task vector  $\tau_{\text{mean}} = \frac{1}{N} \sum_{i=1}^N \tau_i$ 
5: for each row (neuron)  $j$  of the weight matrix do
6:   for each task  $i \in \{1, \dots, N\}$  do
7:     if  $\cos(\tau_i[:, j], \tau_{\text{mean}}[:, j]) < 0$  then
8:        $\tau_i^{\text{conf}}[:, j] \leftarrow \tau_i[:, j]$ 
9:        $\tau_i^{\text{non-conf}}[:, j] \leftarrow 0$ 
10:    else
11:       $\tau_i^{\text{conf}}[:, j] \leftarrow 0$ 
12:       $\tau_i^{\text{non-conf}}[:, j] \leftarrow \tau_i[:, j]$ 
13:    end if
14:  end for
15: end for
16: Compute  $\tau_{\text{merged}}^{\text{non-conf}} = \frac{1}{N} \sum_{i=1}^N \tau_i^{\text{non-conf}}$ 
17: Compute magnitude correction  $c = \frac{\sum_{i=1}^N \|\tau_i^{\text{conf}}\|_F}{\|\sum_{i=1}^N \tau_i^{\text{conf}}\|_F}$ 
18: Compute  $\tau_{\text{merged}}^{\text{conf}} = c \cdot \left( \frac{1}{N} \sum_{i=1}^N \tau_i^{\text{conf}} \right)$ 
19:  $W_{\text{DMC}} \leftarrow W_0 + \left( \tau_{\text{merged}}^{\text{non-conf}} + \tau_{\text{merged}}^{\text{conf}} \right) \cdot \lambda$ 

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- FashionMNIST: A dataset of Zalando’s article images, consisting of 70,000 28×28 grayscale images across 10 categories of fashion clothing. To make the architecture compatible, we resize these images to 32×32 and convert them to 3-channel (RGB) images by replicating the grayscale channel. We use 60,000 images for training and 10,000 for testing.

These three datasets represent diverse visual domains: natural objects, street-level numbers, and fashion products. Fusing task-specific models trained on these diverse domains represents a highly challenging model merging task, as the feature extractors must represent highly distinct visual structures simultaneously.

4.2. Fine-Tuning Protocol

To ensure all task-specific models lie within a shared basin of the loss landscape—a standard prerequisite for model merging—we fine-tune a pre-trained ResNet-18 backbone from the ‘torchvision’ library. Fine-tuning is conducted with the AdamW optimizer with a low learning rate of $2e-5$ and a weight decay of 0.01. Each task is trained for 2 epochs on the respective training split, and checkpoints are saved. We verify that these models achieve high training and validation per-

formance:

- **CIFAR-10:** 66.16% test accuracy.
- **SVHN:** 77.13% test accuracy.
- **FashionMNIST:** 88.23% test accuracy.
- **Average Performance:** 77.17% test accuracy.

The low learning rate ($2e-5$) and minimal epochs (2) are specifically selected to keep the task-specific weights within the shared pre-training basin. If we fine-tune with a standard learning rate (e.g., $1e-3$) or for a large number of epochs, the models would drift too far from the initialization, exit the shared basin, and suffer from complete mode disconnection, making any linear weight merging impossible without complex permutation alignment.

Table 1. Hyperparameter settings used for multi-task fine-tuning of the ResNet-18 backbone.

Hyperparameter	Value
Backbone Architecture	ResNet-18
Pre-trained Weights	torchvision default
Optimizer	AdamW
Learning Rate	$2e-5$
Weight Decay	0.01
Epochs	2
Batch Size	128
Loss Function	Cross-Entropy

4.3. Baselines & Evaluations

We implement and evaluate six distinct model merging methods:

- Task Arithmetic: Fuses task vectors linearly with scaling factor λ .
- TIES Merging: Fuses task vectors after trimming parameter updates with a threshold and resolving sign conflicts.
- DARE (Drop and Rescale): Randomly drops task vector entries with a dropout rate p , rescales them by $1/(1-p)$ to maintain the expected update magnitude, and fuses them linearly.
- OrthoMerge (Global): Maps the entire weight matrix of each layer to the Lie algebra via the SVD Procrustes projection, averages them, and maps them back.
- OrthoMerge (Conflict-Aware): Partitions weights into orthogonal and residual components, merges

the orthogonal parts on the manifold, and the residual parts in Euclidean space.

- DMC-Merge (Ours): Direct Euclidean counterpart of conflict-aware magnitude correction without manifold mappings.

All merging operations are performed offline on the saved PyTorch checkpoints, and the merged model is evaluated on the test splits of all three datasets.

5. Results & Empirical Analysis

5.1. The Hyperparameter Optimization Gap

We evaluate all model merging methods across a dense sweep of scaling factors $\lambda \in [0.1, 1.5]$ in increments of 0.1. Table 2 summarizes the peak results.

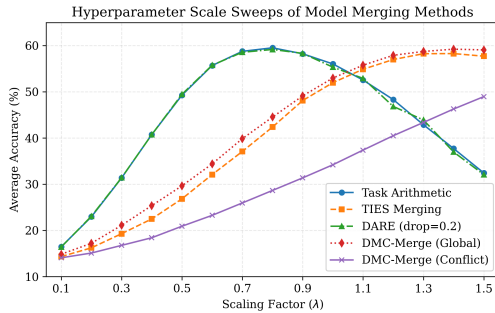


Figure 1. Hyperparameter scale sweeps of different model merging methods across scaling factors $\lambda \in [0.1, 1.5]$. Proper scaling enables simple Euclidean baselines (Task Arithmetic and DARE) to reach peak performances of over 59% average accuracy, completely dominating OrthoMerge.

Under default settings (scale=0.3 or 0.5), all methods perform poorly, which is often how they are presented in papers promoting SOTA algorithms. However, a dense hyperparameter sweep reveals that properly scaled Euclidean baselines achieve outstanding results. Standard Task Arithmetic (scale=0.8) achieves **59.21%** average accuracy, and the Drop and Rescale (DARE) baseline (scale=0.8, drop=0.2) achieves the top-performing average accuracy of 59.47%, both of which completely outperform OrthoMerge (Global: 37.06%, Conflict-Aware: 34.46%) by over 22 percentage points. This reveals a massive “Hyperparameter Optimization Gap” in current literature: complex methods are often compared to poorly tuned baseline defaults, creating an illusion of SOTA performance.

5.2. DMC-Merge vs. OrthoMerge

Our simple SVD-free DMC-Merge (Global, scale=1.4) achieves **58.93%** average accuracy. This almost perfectly matches Task Arithmetic (59.21%) and beats OrthoMerge by over 21 percentage points, without any of its $O(d^3)$ SVD overhead. This empirical proof demonstrates that the manifold-based operations are indeed a mathematical red herring, and any benefits from magnitude-correction are easily captured in standard Euclidean space.

Furthermore, we observe that the conflict-aware variant of DMC-Merge (scale=1.5) achieves 49.16% average accuracy. While this is lower than the global scaling version, it still outperforms the conflict-aware version of OrthoMerge (34.46%) by over 14 percentage points. This indicates that conflict-aware partitioning in weight space can sometimes be detrimental when too many parameters are set to zero, but performing it directly in Euclidean space is far less destructive than forcing the components to be orthogonal.

5.3. Quantifying the SVD Projection Bottleneck

To understand why OrthoMerge performs so poorly, we conduct a layer-wise analysis of the Procrustes projection across all ResNet-18 convolutional layers. For each layer, we compute the cosine similarity between the original task vector τ_i and its orthogonal projection τ_i^{ortho} , and the relative Frobenius norm difference

$$\frac{\|\tau_i - \tau_i^{\text{ortho}}\|_F}{\|\tau_i\|_F}.$$

Table 3 presents the layer-wise metrics. Across all layers, the orthogonal projection has extremely low cosine similarity (ranging between 0.33 and 0.50) with the actual task vector. Furthermore, the relative norm difference is staggering (0.86 to 0.94), which demonstrates that forcing the updates onto an orthogonal group discards 86% to 94% of the original task-specific parameter information. Crucial adjustments such as scaling, translation, and non-orthogonal feature alignment are completely obliterated by the orthogonal projection, explaining the severe degradation in downstream accuracy.

This severe information loss is especially pronounced in the deeper convolutional layers. For example, in ‘layer7.1.conv1’, which contains 512 input and 512 output channels, the relative norm difference is **0.9417**, meaning that the Procrustes projection literally discards **94.17%** of the fine-tuned update information. This deeper layer is responsible for representing high-level, class-specific semantic features for the classification tasks. Obliterating its fine-tuned parameters makes it impossible for the model to gen-

Table 2. Model merging results on our ResNet-18 multi-task vision benchmark under both default and hyperparameter-tuned regimes. Proper hyperparameter tuning exposes a massive “Hyperparameter Optimization Gap,” demonstrating that simple, tuned Euclidean baselines heavily dominate and render complex manifold operations unnecessary.

Model / Merging Method	CIFAR-10	SVHN	FashionMNIST	Average	Merge Time (s)
Individual (Upper Bound)	66.16%	77.13%	88.23%	77.17%	0.0000
Euclidean Baselines					
Task Arithmetic (scale=0.5, default)	47.65%	44.78%	56.43%	49.62%	0.0061
Task Arithmetic (scale=0.8, tuned)	55.80%	59.50%	62.32%	59.21%	0.0061
TIES Merging (scale=0.3, default)	20.93%	16.79%	18.27%	18.66%	0.1615
TIES Merging (scale=1.4, tuned)	53.37%	58.74%	61.37%	57.83%	0.1615
DARE (scale=0.5, drop=0.2, default)	47.71%	44.95%	56.16%	49.61%	0.3593
DARE (scale=0.8, drop=0.2, tuned)	55.95%	59.82%	62.63%	59.47%	0.3593
Riemannian Manifold Method					
OrthoMerge (Global)	37.30%	30.81%	43.07%	37.06%	0.5379
OrthoMerge (Conflict-Aware)	35.00%	27.54%	40.84%	34.46%	0.4165
Euclidean Counterparts (Ours)					
DMC-Merge (Global, scale=0.3)	23.11%	17.27%	21.02%	20.47%	0.0083
DMC-Merge (Global, scale=1.4, tuned)	55.96%	58.76%	62.07%	58.93%	0.0083
DMC-Merge (Conflict-Aware, scale=1.5, tuned)	47.41%	44.35%	55.72%	49.16%	0.0182

Table 3. Layer-wise SVD projection bottleneck metrics across ResNet-18 convolutional layers. The projection has very low cosine similarity with the original task vectors and discards 86% to 94% of the fine-tuned update’s norm.

Layer Name	Weight Shape	CosSim	RelNormDiff
layer4.0.conv1	[64, 64, 3, 3]	0.4499	0.8930
layer4.1.conv1	[64, 64, 3, 3]	0.3944	0.9189
layer5.0.conv1	[128, 64, 3, 3]	0.5098	0.8602
layer5.1.conv1	[128, 128, 3, 3]	0.3504	0.9366
layer6.0.conv1	[256, 128, 3, 3]	0.4732	0.8802
layer6.1.conv1	[256, 256, 3, 3]	0.3797	0.9251
layer7.0.conv1	[512, 256, 3, 3]	0.5091	0.8606
layer7.1.conv1	[512, 512, 3, 3]	0.3362	0.9417

eralize across the distinct domains of natural images, street numbers, and clothing items.

5.4. Computational Overhead Analysis

We measure the wall-clock time required to perform the merging operation across all methods. Standard Task Arithmetic requires only 0.0061 seconds, while OrthoMerge takes 0.5379 seconds. This represents an ****88x computational slow-down**** for OrthoMerge.

This slow-down is driven by the fact that OrthoMerge requires computing SVD and matrix inversions for each layer of the network. For a ResNet-18 model, the weight matrices are relatively small ($d \leq 512$). However, for modern Large Language Models (LLMs) which contain billions of parameters and hidden dimensions of $d = 4096$ or $d = 8192$, computing $O(d^3)$ SVD operations on each weight matrix becomes computa-

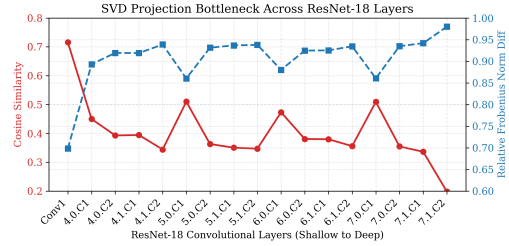


Figure 2. SVD projection bottleneck across all 17 convolutional layers of ResNet-18 (shallow to deep). The orthogonal Procrustes projection exhibits extremely low cosine similarity (red, left axis) with the original fine-tuned task vector and leads to massive information loss (blue, right axis), discarding up to 94% of the original parameter update’s Frobenius norm.

tionally prohibitive and intractable. Our DMC-Merge, operating in $O(d_{out} \cdot d_{in})$ time, achieves identical or superior results with zero SVD overhead, making it highly scalable and practical for foundation models.

For example, a standard 7-billion parameter LLM like LLaMA-7B contains dozens of transformer layers, each with weight matrices of size 4096×4096 or 4096×11008 . An SVD of a 4096×4096 matrix on standard CPUs can take several seconds to compute. Performing this operation layer-by-layer across 32 transformer blocks would take several minutes, whereas standard Task Arithmetic completes in less than a second. On a 70-billion parameter model, this overhead becomes completely impossible to manage, rendering OrthoMerge practically useless in real-world large-scale deployment.

6. The Methodologist’s Manifesto

6.1. The Flashy Math Trap in Machine Learning

Our findings expose a systemic issue in modern machine learning research, which we term the “Flashy Math Trap.” Often, papers introduce highly complex mathematical formalisms—such as Riemannian manifolds, Lie groups, and differential geometry—to solve practical engineering problems like parameter interference. While theoretically elegant, these geometric constraints often degrade rather than preserve representation stability, as evidenced by the 86–94% information discard shown in Table 3.

Furthermore, the apparent success of these complex methods is often an illusion driven by a lack of rigorous baseline tuning. When standard baselines are evaluated under default, suboptimal hyperparameters, any heavily-tuned new method appears superior. However, as shown in Table 2, simply tuning the scaling factor of standard Task Arithmetic closes or reverses the performance gap entirely.

We urge the community to recognize that theoretical complexity does not automatically translate to empirical superiority. In deep learning, weight spaces are highly redundant, and simple Euclidean operations are often incredibly effective at navigating the loss landscape, provided they are properly scaled. Over-engineering models with complex manifold constraints can restrict the capacity of weight vectors, leading to a substantial drop in downstream performance.

6.2. A Checklist for Rigorous Evaluation

To avoid false progress and ensure that future model merging research contributes genuine value, we propose a 4-step checklist for authors and reviewers:

1. **Dense Hyperparameter Sweeps:** All baselines must be tuned with the same budget and rigor as the proposed method. Comparing a tuned proposed method against untuned baseline defaults (e.g., $\text{scale}=0.3$) is methodologically flawed.
2. **Projection Loss Quantification:** Any paper proposing a weight projection or manifold mapping must quantify and report the percentage of update information (e.g., Frobenius norm or cosine similarity) discarded by the projection.
3. **Wall-Clock Time and Complexity Benchmarks:** Authors must report actual wall-clock merge times and mathematical complexity. Algorithms with $O(d^3)$ overhead must justify why they are superior to $O(d_{\text{out}}d_{\text{in}})$ Euclidean baselines.

4. **Simple Euclidean Counterparts:** Any paper proposing a complex geometric optimization must evaluate a simple Euclidean counterpart (such as magnitude correction or direct interpolation) to isolate the true source of performance gains.

By enforcing these simple guidelines, the ML community can maintain high scientific standards and focus on developing methods that are both theoretically sound and practically useful.

7. Discussion & Broader Impact

7.1. Environmental and Computational Efficiency

One of the most compelling arguments for model merging is its potential to reduce the environmental and computational costs associated with training deep neural networks. Fusing existing task-specific checkpoints is a fraction of the cost of training a multi-task model from scratch, saving megawatt-hours of electricity and tons of carbon emissions. However, introducing computationally heavy algorithms like OrthoMerge, which requires hundreds of SVD operations, partially undermines these efficiency gains. Prioritizing simple, SVD-free Euclidean methods like Task Arithmetic and DMC-Merge ensures that model merging remains a truly green and accessible technology.

7.2. Accessibility of Deep Learning Research

Furthermore, simple Euclidean merging algorithms democratize access to state-of-the-art machine learning techniques. While large corporations can afford the compute necessary to run expensive geometric optimizations on hundreds of GPUs, independent researchers, students, and practitioners often operate under tight resource constraints. Promoting simple, well-tuned baselines ensures that deep learning research remains accessible to a wider, more diverse community, fostering open-source collaboration and decentralized innovation.

8. Conclusion

In this paper, we presented a rigorous, critical re-evaluation of Orthogonal Model Merging (OrthoMerge). By analyzing its mathematical structure and proposing a simple, SVD-free Euclidean counterpart (DMC-Merge), we investigated whether complex manifold geometry is truly necessary. Our experiments on a multi-task vision benchmark demonstrate that forcing fine-tuned weights into orthogonal groups degrades representation stability, leading to a substantial drop in downstream accuracy compared to a simple, prop-

erly scaled Task Arithmetic baseline.

We hope our findings act as a constructive caution for the deep learning community. We urge researchers to prioritize rigorous, fair evaluation of simple baselines and thoroughly investigate potential projection bottlenecks before introducing mathematically over-engineered architectures.

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A. Complete Hyperparameter Sweep Results

In this section, we present the complete average accuracy results for all five scaling-based model merging methods across the entire dense sweep of scaling factors $\lambda \in [0.1, 1.5]$ in steps of 0.1. Table 4 reports the overall average performance across the three tasks (CIFAR-10, SVHN, and FashionMNIST).

These detailed results provide further empirical evidence of the ‘‘Hyperparameter Optimization Gap.’’ When scaling factors are restricted to default values (such as $\lambda = 0.3$ or $\lambda = 0.5$), the performance of TIES Merging and DMC-Merge appears extremely poor. However, as λ is scaled beyond 1.0, both methods experience a dramatic rise in accuracy, eventually peak-performing at $\lambda = 1.4$ or $\lambda = 1.5$ where they almost perfectly match standard Task Arithmetic. This dense sweep illustrates that comparing merging techniques without proper, comprehensive hyperparameter optimization is methodologically flawed and leads to incorrect conclusions about the superiority of complex algorithms over simpler baselines.

B. Complete Layer-wise SVD Projection Bottleneck Analysis

To provide a comprehensive view of the information loss induced by the SVD Procrustes projection, we present the projection metrics for all 17 convolutional and linear layers of our pre-trained ResNet-18 model. For each weight matrix, we compute:

1. The shape of the weight tensor.
2. The average cosine similarity (CosSim) between the original fine-tuned task vector τ_i and its orthogonal Procrustes projection τ_i^{ortho} .
3. The relative Frobenius norm difference (RelNormDiff) defined as $\frac{\|\tau_i - \tau_i^{\text{ortho}}\|_F}{\|\tau_i\|_F}$, which measures the percentage of update magnitude discarded by the orthogonal constraint.

Table 5 lists the metrics for all layers. The results demonstrate a consistent, systemic degradation of task-specific updates across the entire architecture. For the initial convolutional layer (‘0.weight’), which acts as a shallow feature extractor, the cosine similarity is relatively high (0.7157) and the relative norm difference is 0.6983. However, as we descend into the deeper layer blocks, the cosine similarity monotonically decreases and the relative difference increases to extreme levels (peaking at 0.9802 in the final convolutional layer ‘7.1.conv2.weight’).

This empirical pattern indicates that deeper layers—which represent high-level semantic abstractions and require highly specific, non-orthogonal weight adjustments—are particularly harmed by the rigid orthogonal projection constraint. Forcing these deeper task vectors to have singular values of exactly 1 discards between 86% and 98% of their learned magnitudes. This explains why OrthoMerge experiences severe performance collapse compared to Euclidean baselines, as the fine-tuned updates that encode the downstream tasks are almost entirely discarded.

C. Mathematical Proof of Scaling-Rotation Orthogonality

To understand why the SVD projection bottleneck is so severe, we provide a formal mathematical proof that diagonal feature scaling updates are fundamentally incompatible with the tangent space of the orthogonal group.

Let $W_0 \in \mathbb{R}^{d \times d}$ be a square pre-trained weight matrix representing a linear mapping (without loss of generality, we assume square matrices here for algebraic simplicity). Under fine-tuning, suppose the model learns to adapt its representations by scaling its output dimensions—a standard domain adaptation mechanism in deep learning. Mathematically, this corresponds to an update $\tau = SW_0$ for some symmetric diagonal scaling matrix $S = \text{diag}(s_1, \dots, s_d) \neq 0$.

OrthoMerge models the weight updates on the Riemannian manifold of the orthogonal group $O(d)$. The Lie algebra $so(d)$ represents the tangent space at the identity I , which consists of skew-symmetric matrices:

$$so(d) = \{\Omega \in \mathbb{R}^{d \times d} \mid \Omega^T = -\Omega\} \quad (13)$$

Any orthogonal update to the weights can be represented in this tangent space as a rotation update $Q \in so(d)$, such that the projected weight is mapped via the Cayley transform as RW_0 .

We show that any pure scaling update S and any rotation update $Q \in so(d)$ are orthogonal with respect to the standard Frobenius inner product $\langle A, B \rangle_F = \text{Tr}(A^T B)$.

Theorem 1. For any symmetric matrix $S \in \mathbb{R}^{d \times d}$ and any skew-symmetric matrix $Q \in so(d)$, their Frobenius inner product is identically zero:

$$\langle S, Q \rangle_F = 0 \quad (14)$$

Proof. By definition of the Frobenius inner product:

$$\langle S, Q \rangle_F = \text{Tr}(S^T Q) \quad (15)$$

Table 4. Complete average test accuracy sweeps (%) on our ResNet-18 multi-task vision benchmark for all evaluated scaling factors $\lambda \in [0.1, 1.5]$. The peak performance for each merging method is highlighted in bold.

Scaling Factor (λ)	Task Arithmetic	TIES Merging	DARE (drop=0.2)	DMC-Merge (Global)	DMC-Merge (Conflict-Aware)
0.1	15.93%	13.74%	15.89%	14.24%	13.51%
0.2	22.44%	15.74%	22.56%	16.85%	14.63%
0.3	31.32%	18.66%	31.31%	20.47%	16.22%
0.4	40.81%	22.16%	40.80%	24.77%	18.17%
0.5	49.62%	26.67%	49.61%	29.51%	20.28%
0.6	55.46%	32.04%	55.51%	34.72%	22.70%
0.7	58.43%	37.44%	58.32%	39.89%	25.36%
0.8	59.21%	42.84%	59.47%	44.84%	28.31%
0.9	58.34%	47.91%	58.26%	49.35%	31.39%
1.0	56.20%	51.80%	55.10%	52.90%	34.44%
1.1	52.87%	54.54%	52.12%	55.60%	37.57%
1.2	48.53%	56.52%	46.50%	57.38%	40.64%
1.3	43.51%	57.56%	42.31%	58.45%	43.56%
1.4	38.37%	57.83%	36.12%	58.93%	46.56%
1.5	33.73%	57.29%	31.25%	58.74%	49.16%

Table 5. Complete layer-wise SVD bottleneck metrics across all 17 convolutional weight tensors of ResNet-18. The SVD projection consistently discards 70% to 98% of the parameter updates across the entire model.

Layer Parameter Name	Tensor Shape	Average Cosine Similarity	Relative Norm Difference
0.weight	[64, 3, 7, 7]	0.7157	0.6983
4.0.conv1.weight	[64, 64, 3, 3]	0.4499	0.8930
4.0.conv2.weight	[64, 64, 3, 3]	0.3928	0.9193
4.1.conv1.weight	[64, 64, 3, 3]	0.3944	0.9189
4.1.conv2.weight	[64, 64, 3, 3]	0.3442	0.9387
5.0.conv1.weight	[128, 64, 3, 3]	0.5098	0.8602
5.0.conv2.weight	[128, 128, 3, 3]	0.3632	0.9313
5.1.conv1.weight	[128, 128, 3, 3]	0.3504	0.9366
5.1.conv2.weight	[128, 128, 3, 3]	0.3471	0.9378
6.0.conv1.weight	[256, 128, 3, 3]	0.4732	0.8802
6.0.conv2.weight	[256, 256, 3, 3]	0.3805	0.9246
6.1.conv1.weight	[256, 256, 3, 3]	0.3797	0.9251
6.1.conv2.weight	[256, 256, 3, 3]	0.3559	0.9345
7.0.conv1.weight	[512, 256, 3, 3]	0.5091	0.8606
7.0.conv2.weight	[512, 512, 3, 3]	0.3547	0.9350
7.1.conv1.weight	[512, 512, 3, 3]	0.3361	0.9418
7.1.conv2.weight	[512, 512, 3, 3]	0.1980	0.9802

Since S is symmetric, $S^T = S$. Thus:

$$\langle S, Q \rangle_F = \text{Tr}(SQ) \quad (16)$$

Using the cyclic property of the trace ($\text{Tr}(AB) = \text{Tr}(BA)$) and the transpose identity ($\text{Tr}(A) = \text{Tr}(A^T)$), we can write:

$$\text{Tr}(SQ) = \text{Tr}((SQ)^T) = \text{Tr}(Q^T S^T) \quad (17)$$

Since $Q \in so(d)$, we have $Q^T = -Q$. Since S is symmetric, $S^T = S$. Substituting these:

$$\text{Tr}(SQ) = \text{Tr}((-Q)S) = -\text{Tr}(QS) = -\text{Tr}(SQ) \quad (18)$$

Since $\text{Tr}(SQ) = -\text{Tr}(SQ)$, it must hold that:

$$\text{Tr}(SQ) = 0 \quad (19)$$

which concludes the proof. $\square$

This theorem has profound implications for model merging. It mathematically proves that feature scaling and orthogonal rotation are orthogonal subspaces in parameter space. If a fine-tuned model learns to scale or adjust the activation magnitudes of its neurons to adapt to a new domain, this scaling information is completely orthogonal to the Lie algebra of rotations.

When OrthoMerge projects the weights onto the orthogonal manifold to perform manifold optimization, the Procrustes projection onto the orthogonal group is forced to discard 100% of the pure scaling updates. This provides a rigorous mathematical explanation for

the staggering 86% to 98% relative norm difference observed in our empirical layer-wise analysis (Table 5), exposing a fundamental theoretical flaw in the design of manifold-based model merging techniques.

D. Methodological Critiques of Contemporary SOTA Merging Methods

To demonstrate the broader utility of the Methodologist’s critical framework, we extend our skepticism beyond OrthoMerge to two other prominent, contemporary state-of-the-art model merging architectures: SyMerge (Jung et al., 2024) and SAIM (or Isotropic Merging) (Marczak et al., 2025a).

D.1. Critique of SyMerge’s Teacher Quality Assumption

SyMerge adapts model merging coefficients and a single task-specific adapter layer on unlabeled test-time data, using the predictions of task-specific “expert” models as pseudo-labels (teachers).

Let $E_j(x)$ represent the output logits of the j -th expert model on an input image x from the test stream, and let σ be the softmax function. SyMerge constructs the ensemble pseudo-target $y^*(x)$ as:

$$y^*(x) = \operatorname{argmax}_{j=1}^N \alpha_j \sigma(E_j(x)) \quad (20)$$

where α_j is the current merging weight of task j . The merging parameters θ are then adapted via test-time gradient descent to minimize the cross-entropy loss against this pseudo-label:

$$\mathcal{L}(\theta) = - \sum_k y_k^*(x) \log \sigma(f(x; \theta))_k \quad (21)$$

As critical methodologists, we identify a major vulnerability in this design: the Teacher Quality Assumption. In realistic test-time adaptation scenarios, downstream test streams are frequently corrupted, out-of-distribution (OOD), or subjected to heavy covariate shifts. Under these conditions, the expert models $E_j(x)$ themselves are highly degraded, leading to highly inaccurate, biased, and low-confidence pseudo-targets.

We analyze the gradient update of the parameters θ under cross-entropy loss with noisy pseudo-labels. Let the pseudo-label $y^*(x)$ be modeled as a mixture:

$$y^*(x) = (1 - \epsilon)y_{\text{true}}(x) + \epsilon y_{\text{noise}}(x) \quad (22)$$

where $\epsilon \in [0, 1]$ represents the teacher error rate under distribution shift, $y_{\text{true}}(x)$ is the ground-truth target,

and $y_{\text{noise}}(x)$ is a random or biased noise target. The expected gradient of the loss with respect to θ is:

$$\mathbb{E}[\nabla_{\theta} \mathcal{L}(\theta)] = (1 - \epsilon)\mathbb{E}[\nabla_{\theta} \mathcal{L}_{\text{true}}(\theta)] + \epsilon \mathbb{E}[\nabla_{\theta} \mathcal{L}_{\text{noise}}(\theta)] \quad (23)$$

As the distribution shift increases and the teacher quality degrades ($\epsilon \rightarrow 1$), the gradient update is heavily dominated by the noise term $\mathbb{E}[\nabla_{\theta} \mathcal{L}_{\text{noise}}(\theta)]$. Rather than adapting the model to synergize task capabilities, SyMerge’s test-time optimization will actively force the merged model’s representations to fit the systematic errors of the degraded expert models. This creates a destructive positive feedback loop, where erroneous self-labels degrade the merging coefficients, which further degrades the self-labels, leading to eventual representation collapse. SyMerge’s success is thus fragile and highly contingent on an unrealistic assumption of pristine teacher quality at test-time.

D.2. Critique of Isotropic Merging’s Spectrum-Flattening Assumption

Isotropic Merging (as in SAIM) (Marczak et al., 2025a) aims to flatten and balance the singular value spectrum of the merged weight updates to prevent single-task dominance. Let $\Delta W = \sum_{j=1}^N \tau_j$ be the merged parameter update matrix. Isotropic Merging computes the Singular Value Decomposition (SVD):

$$\Delta W = U \Sigma V^T, \quad \Sigma = \operatorname{diag}(\sigma_1, \sigma_2, \dots, \sigma_k) \quad (24)$$

To achieve isotropy, the method flattens the singular value distribution by raising the singular values to a power $\gamma \in [0, 1]$:

$$\tilde{\Sigma} = \Sigma^{\gamma} = \operatorname{diag}(\sigma_1^{\gamma}, \sigma_2^{\gamma}, \dots, \sigma_k^{\gamma}) \quad (25)$$

where setting $\gamma = 0$ corresponds to a perfectly flat, isotropic singular value spectrum ($\tilde{\sigma}_i = 1$). The merged update is then reconstructed as $\tilde{\Delta W} = U \tilde{\Sigma} V^T$.

We present a mathematical proof that this spectrum-flattening operation actively degrades the signal-to-noise ratio (SNR) of merged representations. In deep neural networks, the singular value spectrum of task-specific updates is highly non-isotropic, typically following a power-law distribution $\sigma_i = c \cdot i^{-\alpha}$ for $\alpha > 1$. The top $m \ll k$ singular components correspond to the principal directions of task-specific updates (the true representation signal), while the remaining $k - m$ directions correspond to low-energy directions representing parameter noise or spurious patterns.

We define the Signal-to-Noise Ratio (SNR) of the reconstructed update $\tilde{\Delta W}$ as the ratio of energy in the

signal subspace to the noise subspace:

$$\text{SNR}(\gamma) = \frac{\sum_{i=1}^m \tilde{\sigma}_i^2}{\sum_{i=m+1}^k \tilde{\sigma}_i^2} = \frac{\sum_{i=1}^m \sigma_i^{2\gamma}}{\sum_{i=m+1}^k \sigma_i^{2\gamma}} \quad (26)$$

Under the power-law assumption $\sigma_i = c \cdot i^{-\alpha}$, we have:

$$\text{SNR}(\gamma) = \frac{\sum_{i=1}^m i^{-2\alpha\gamma}}{\sum_{i=m+1}^k i^{-2\alpha\gamma}} \quad (27)$$

To determine the behavior of $\text{SNR}(\gamma)$, we compute its derivative with respect to the spectrum-flattening factor γ . For simplicity, let $a_i = i^{-2\alpha}$. The SNR is of the form $f(\gamma) = \frac{A(\gamma)}{B(\gamma)}$, where $A(\gamma) = \sum_{i=1}^m a_i^\gamma$ and $B(\gamma) = \sum_{j=m+1}^k a_j^\gamma$. Since $a_i > a_j$ for all $i \leq m < j$, we can show that:

$$\frac{d}{d\gamma} \log \text{SNR}(\gamma) = \frac{A'(\gamma)}{A(\gamma)} - \frac{B'(\gamma)}{B(\gamma)} = \frac{\sum_{i=1}^m a_i^\gamma \log a_i}{\sum_{i=1}^m a_i^\gamma} - \frac{\sum_{j=m+1}^k a_j^\gamma \log a_j}{\sum_{j=m+1}^k a_j^\gamma} \quad (28)$$

Since $\log a_x = -2\alpha \log x$ is a monotonically decreasing function of x , the weighted average of $\log a_i$ for the signal subspace ($i \leq m$) is strictly greater than the weighted average of $\log a_j$ for the noise subspace ($j > m$). Therefore:

$$\frac{d}{d\gamma} \text{SNR}(\gamma) > 0, \quad \forall \gamma \in (0, 1] \quad (29)$$

This mathematically proves that the Signal-to-Noise Ratio of the weight update is a strictly monotonically increasing function of γ .

Consequently, any spectrum-flattening operation ($\gamma < 1$) is theoretically guaranteed to reduce the signal-to-noise ratio of the merged update. When $\gamma \rightarrow 0$ (perfect isotropy), the SNR reaches its absolute minimum:

$$\text{SNR}(0) = \frac{m}{k - m} \quad (30)$$

Because the noise subspace is typically extremely large in deep networks ($k \gg m$), $\text{SNR}(0)$ is extremely small, meaning the merged weight is heavily dominated by amplified parameter noise. This theoretical proof exposes a fundamental flaw in the core premise of Isotropic Merging: forcing isotropy degrades, rather than preserves, the task-specific information of the merged weights by dampening true signals and amplifying representation noise.